

H524 Introduction to Biostatistics

Week 9 Lab: Correlation and Simple Linear Regression

Fall 2025

Format: Online

Tools Required: R, RStudio

Estimated Time: 90 minutes

1 Learning Objectives

By the end of this lab, you will be able to:

1. Calculate and interpret Pearson and Spearman correlation coefficients
2. Test the significance of correlation coefficients
3. Create scatter plots with informative labels and regression lines
4. Fit simple linear regression models using `lm()`
5. Interpret regression output (coefficients, R^2 , p -values)
6. Check regression assumptions using diagnostic plots
7. Make predictions using regression models
8. Calculate confidence and prediction intervals
9. [Use AI tools to help with regression analysis and interpretation](#)

2 Part 1: Correlation Analysis

Correlation measures the strength and direction of the linear relationship between two continuous variables.

2.1 Exercise 1.1: Calculating Correlation

Scenario: We have data on age and systolic blood pressure for 15 adults.

```
# Age and systolic blood pressure data
age <- c(35, 38, 40, 42, 45, 47, 48, 50, 52, 55, 58, 60, 62, 65, 68)
sbp <- c(118, 122, 125, 128, 130, 132, 135, 136, 138, 140, 145, 148,
        150, 155, 158)

# Calculate Pearson correlation coefficient
cor_result <- cor.test(age, sbp)
print(cor_result)

# Extract key information
cat("\nPearson correlation coefficient: r =", round(cor_result$estimate, 3), "\n")
```

```

cat("95% Confidence Interval:", round(cor_result$conf.int, 3), "\n")
cat("t-statistic:", round(cor_result$statistic, 3), "\n")
cat("p-value:", format.pval(cor_result$p.value, digits=3), "\n")

# Interpretation
if(cor_result$p.value < 0.05) {
  cat("\nConclusion: There is significant evidence of correlation (p < 0.05)\n")
} else {
  cat("\nConclusion: No significant evidence of correlation (p >= 0.05)\n")
}

```

Questions:

- What is the value of the correlation coefficient? Is it positive or negative?
- What does this tell us about the relationship between age and blood pressure?
- Is the correlation statistically significant?

2.2 Exercise 1.2: Visualizing Correlation

Always visualize your data before computing statistics!

```

# Create scatter plot
plot(age, sbp,
     main = "Systolic Blood Pressure vs. Age",
     xlab = "Age (years)",
     ylab = "Systolic BP (mmHg)",
     pch = 19,
     col = "blue",
     cex = 1.2)

# Add grid for easier reading
grid()

# Add correlation coefficient to plot
text(40, 155,
     paste("r =", round(cor(age, sbp), 3)),
     pos = 4, cex = 1.2, col = "red")

# Optionally, add a lowess smoother to see the trend
lines(lowess(age, sbp), col = "red", lwd = 2)

```

Observation: Does the scatter plot support the correlation coefficient you calculated? Do you see a linear trend?

2.3 Exercise 1.3: Spearman Correlation

Spearman correlation is based on ranks and is more robust to outliers.

```
# Calculate Spearman correlation
spearman_result <- cor.test(age, sbp, method = "spearman")
print(spearman_result)

# Compare Pearson and Spearman
pearson_r <- cor(age, sbp, method = "pearson")
spearman_r <- cor(age, sbp, method = "spearman")

cat("\nPearson r:", round(pearson_r, 3), "\n")
cat("Spearman rho:", round(spearman_r, 3), "\n")
cat("Difference:", round(abs(pearson_r - spearman_r), 3), "\n")
```

Question: Why are Pearson and Spearman correlations similar in this case? When would they differ substantially?

2.4 Exercise 1.4: Effect of Outliers on Correlation

Let's see how outliers affect correlation.

```
# Create data with an outlier
age_outlier <- c(age, 45)
sbp_outlier <- c(sbp, 95) # Unusually low BP for age 45

# Calculate correlations with and without outlier
r_original <- cor(age, sbp)
r_outlier <- cor(age_outlier, sbp_outlier)

cat("Original correlation:", round(r_original, 3), "\n")
cat("With outlier:", round(r_outlier, 3), "\n")
cat("Change:", round(r_outlier - r_original, 3), "\n")

# Visualize the effect
par(mfrow = c(1, 2))

# Original data
plot(age, sbp, main = "Original Data",
     xlab = "Age", ylab = "SBP", pch = 19, col = "blue")
text(40, 155, paste("r =", round(r_original, 3)), col = "red")

# With outlier
plot(age_outlier, sbp_outlier, main = "With Outlier",
     xlab = "Age", ylab = "SBP", pch = 19, col = "blue")
points(45, 95, col = "red", pch = 19, cex = 2) # Highlight outlier
text(40, 155, paste("r =", round(r_outlier, 3)), col = "red")
```

```
par(mfrow = c(1, 1))
```

Key lesson: Always examine your data visually. A single outlier can dramatically change the correlation!

3 Part 2: Simple Linear Regression

Regression allows us to model the relationship and make predictions.

3.1 Exercise 2.1: Fitting a Regression Model

```
# Fit simple linear regression: sbp ~ age
model <- lm(sbp ~ age)

# Display full summary
summary(model)

# Extract key components
coef_table <- summary(model)$coefficients
r_squared <- summary(model)$r.squared
adj_r_squared <- summary(model)$adj.r.squared
residual_se <- summary(model)$sigma

cat("\n=== Regression Results Summary ===\n")
cat("Intercept (b0):", round(coef(model)[1], 2), "mmHg\n")
cat("Slope (b1):", round(coef(model)[2], 2), "mmHg/year\n")
cat("R-squared:", round(r_squared, 4), "\n")
cat("Adjusted R-squared:", round(adj_r_squared, 4), "\n")
cat("Residual standard error:", round(residual_se, 2), "mmHg\n")

# Regression equation
cat("\nRegression equation:\n")
cat("SBP =", round(coef(model)[1], 2), "+",
    round(coef(model)[2], 2), "* Age\n")
```

Interpretation questions:

- What is the slope? Interpret it in context.
- What is the intercept? Is it meaningful in this context?
- What percentage of variability in SBP is explained by age?
- Is the relationship statistically significant?

3.2 Exercise 2.2: Visualizing the Regression Line

```
# Scatter plot with regression line
plot(age, sbp,
     main = "Systolic BP vs. Age with Regression Line",
     xlab = "Age (years)",
     ylab = "Systolic BP (mmHg)",
     pch = 19,
     col = "blue",
     cex = 1.2)

# Add regression line
abline(model, col = "red", lwd = 2)

# Add confidence band
age_pred <- seq(min(age), max(age), length.out = 100)
conf_interval <- predict(model,
                        newdata = data.frame(age = age_pred),
                        interval = "confidence",
                        level = 0.95)

# Plot confidence bands
lines(age_pred, conf_interval[, "lwr"], col = "gray", lty = 2, lwd = 1.5)
lines(age_pred, conf_interval[, "upr"], col = "gray", lty = 2, lwd = 1.5)

# Add equation to plot
equation <- paste("SBP =", round(coef(model)[1], 1), "+",
                  round(coef(model)[2], 2), "* Age")
text(40, 155, equation, pos = 4, col = "red", cex = 1)

# Add R-squared
text(40, 150, paste("R-squared =", round(r_squared, 3)),
     pos = 4, col = "red", cex = 1)

# Add legend
legend("topleft",
      legend = c("Observed data", "Regression line", "95% CI"),
      col = c("blue", "red", "gray"),
      lty = c(NA, 1, 2),
      pch = c(19, NA, NA),
      lwd = c(NA, 2, 1.5))
```

3.3 Exercise 2.3: Making Predictions

```
# Predict SBP for specific ages
new_ages <- data.frame(age = c(45, 55, 65))

# Point predictions
predictions <- predict(model, newdata = new_ages)
cat("Point predictions:\n")
```

```

for(i in 1:nrow(new_ages)) {
  cat("Age", new_ages$age[i], "years: SBP =",
      round(predictions[i], 1), "mmHg\n")
}

# Confidence intervals for mean response
conf_int <- predict(model, newdata = new_ages,
                    interval = "confidence", level = 0.95)
cat("\n95% Confidence Intervals (for mean SBP):\n")
print(round(conf_int, 1))

# Prediction intervals for individual response
pred_int <- predict(model, newdata = new_ages,
                    interval = "prediction", level = 0.95)
cat("\n95% Prediction Intervals (for individual SBP):\n")
print(round(pred_int, 1))

# Compare widths
cat("\nNotice: Prediction intervals are WIDER than confidence intervals\n")
cat("CI width for age 55:", round(conf_int[2,"upr"] - conf_int[2,"lwr"], 1), "mmHg\n")
cat("PI width for age 55:", round(pred_int[2,"upr"] - pred_int[2,"lwr"], 1), "mmHg\n")

```

Question: Why are prediction intervals wider than confidence intervals?

3.4 Exercise 2.4: Checking Regression Assumptions

Regression requires several assumptions. Let's check them!

```

# Create 2x2 panel of diagnostic plots
par(mfrow = c(2, 2))
plot(model)
par(mfrow = c(1, 1))

# Residuals vs Fitted: checks linearity and constant variance
# Normal Q-Q: checks normality of residuals
# Scale-Location: checks homoscedasticity
# Residuals vs Leverage: checks for influential points

# Additional tests
cat("\n=== Assumption Checks ===\n")

# Test for normality of residuals
shapiro_test <- shapiro.test(residuals(model))
cat("\nShapiro-Wilk test for normality:\n")
cat("W =", round(shapiro_test$statistic, 4),
    ", p-value =", round(shapiro_test$p.value, 4), "\n")
if(shapiro_test$p.value > 0.05) {
  cat("Residuals appear normally distributed (p > 0.05)\n")
}

```

```

} else {
  cat("Evidence against normality (p < 0.05)\n")
}

# Examine residuals
cat("\nResidual summary:\n")
print(summary(residuals(model)))

```

What to look for:

- **Residuals vs Fitted:** Should show random scatter around zero (no pattern)
- **Normal Q-Q:** Points should follow diagonal line (normality)
- **Scale-Location:** Should show random scatter (constant variance)
- **Residuals vs Leverage:** No points with high leverage and large residuals

3.5 Exercise 2.5: ANOVA Table for Regression

```

# ANOVA table
anova_table <- anova(model)
print(anova_table)

# Extract components
ss_total <- sum((sbp - mean(sbp))^2)
ss_regression <- anova_table$'Sum Sq'[1]
ss_residual <- anova_table$'Sum Sq'[2]

cat("\n=== ANOVA Decomposition ===\n")
cat("Total SS:", round(ss_total, 2), "\n")
cat("Regression SS:", round(ss_regression, 2), "\n")
cat("Residual SS:", round(ss_residual, 2), "\n")
cat("R-squared = Regression SS / Total SS =",
    round(ss_regression / ss_total, 4), "\n")

# F-test
f_stat <- anova_table$'F value'[1]
p_value <- anova_table$'Pr(>F)'[1]
cat("\nF-statistic:", round(f_stat, 2), "\n")
cat("p-value:", format.pval(p_value, digits=3), "\n")
cat("\nConclusion: The regression model is",
    ifelse(p_value < 0.05, "significant", "not significant"), "\n")

```

4 Part 3: Real-World Examples

4.1 Exercise 3.1: Birth Weight and Gestational Age

```
# Simulated data based on realistic values
set.seed(524)
gestational_age <- round(runif(50, 32, 42))
birth_weight <- 150 * gestational_age - 2800 + rnorm(50, 0, 250)

# Create data frame
birth_data <- data.frame(weeks = gestational_age, weight = birth_weight)

# Summary
cat("Birth Weight Study\n")
cat("Sample size:", nrow(birth_data), "\n")
cat("Gestational age range:", range(birth_data$weeks), "weeks\n")
cat("Birth weight range:", round(range(birth_data$weight)), "grams\n\n")

# Correlation
cor_birth <- cor.test(birth_data$weeks, birth_data$weight)
cat("Pearson correlation: r =", round(cor_birth$estimate, 3), "\n")
cat("p-value:", format.pval(cor_birth$p.value, digits=3), "\n\n")

# Regression
model_birth <- lm(weight ~ weeks, data = birth_data)
summary(model_birth)

# Plot
plot(birth_data$weeks, birth_data$weight,
     main = "Birth Weight vs. Gestational Age",
     xlab = "Gestational Age (weeks)",
     ylab = "Birth Weight (grams)",
     pch = 19, col = "darkgreen", cex = 1.2)
abline(model_birth, col = "red", lwd = 2)

# Add interpretation
cat("\n=== Clinical Interpretation ===\n")
cat("Slope:", round(coef(model_birth)[2], 1), "grams per week\n")
cat("Each additional week of gestation increases birth weight by",
     round(coef(model_birth)[2], 0), "grams on average\n")
cat("R-squared:", round(summary(model_birth)$r.squared, 3), "\n")
cat("Gestational age explains",
     round(100 * summary(model_birth)$r.squared, 1),
     "% of variability in birth weight\n")
```

Practice: Use the model to predict birth weight for babies at 36, 38, and 40 weeks gestation. Include 95% prediction intervals.

4.2 Exercise 3.2: Exercise and Resting Heart Rate

```
# Simulated data
set.seed(123)
exercise_sessions <- sample(0:7, 30, replace = TRUE)
resting_hr <- 78 - 2.5 * exercise_sessions + rnorm(30, 0, 5)

# Create data frame
hr_data <- data.frame(sessions = exercise_sessions, hr = resting_hr)

# Scatter plot
plot(hr_data$sessions, hr_data$hr,
     main = "Resting Heart Rate vs. Exercise Frequency",
     xlab = "Exercise Sessions per Week",
     ylab = "Resting Heart Rate (bpm)",
     pch = 19, col = "purple", cex = 1.2)

# Correlation
cor_hr <- cor.test(hr_data$sessions, hr_data$hr)
cat("Correlation: r =", round(cor_hr$estimate, 3), "\n")
cat("p-value:", format.pval(cor_hr$p.value, digits=3), "\n\n")

# Regression
model_hr <- lm(hr ~ sessions, data = hr_data)
summary(model_hr)
abline(model_hr, col = "red", lwd = 2)

# Interpretation
cat("\n=== Interpretation ===\n")
cat("Slope:", round(coef(model_hr)[2], 2), "bpm per session\n")
cat("Each additional exercise session per week is associated with",
     abs(round(coef(model_hr)[2], 1)), "bpm decrease in resting heart rate\n")

# Check assumptions
par(mfrow = c(2, 2))
plot(model_hr)
par(mfrow = c(1, 1))
```

5 Part 4: Advanced Topics

5.1 Exercise 4.1: Comparing Two Regression Models

```
# Compare models with different predictors
# Example: Height and weight data
set.seed(456)
height_cm <- rnorm(40, 170, 10)
weight_kg <- 0.8 * height_cm - 70 + rnorm(40, 0, 5)
age_years <- sample(20:60, 40, replace = TRUE)

# Model 1: Weight predicted by height
```

```

model1 <- lm(weight_kg ~ height_cm)

# Model 2: Weight predicted by age
model2 <- lm(weight_kg ~ age_years)

# Compare R-squared
cat("Model 1 (Height): R-squared =", round(summary(model1)$r.squared, 3), "\n")
cat("Model 2 (Age): R-squared =", round(summary(model2)$r.squared, 3), "\n\n")

# Compare AIC (lower is better)
cat("Model 1 (Height): AIC =", round(AIC(model1), 2), "\n")
cat("Model 2 (Age): AIC =", round(AIC(model2), 2), "\n\n")

# Visualize both models
par(mfrow = c(1, 2))
plot(height_cm, weight_kg, main = "Model 1: Height",
      xlab = "Height (cm)", ylab = "Weight (kg)", pch = 19)
abline(model1, col = "red", lwd = 2)
text(160, 100, paste("R =", round(summary(model1)$r.squared, 3)))

plot(age_years, weight_kg, main = "Model 2: Age",
      xlab = "Age (years)", ylab = "Weight (kg)", pch = 19)
abline(model2, col = "blue", lwd = 2)
text(25, 100, paste("R =", round(summary(model2)$r.squared, 3)))
par(mfrow = c(1, 1))

```

5.2 Exercise 4.2: Influence of Sample Size

```

# Demonstrate effect of sample size on inference
set.seed(789)

sample_sizes <- c(10, 30, 100, 300)
results <- data.frame(n = sample_sizes, r = NA, p_value = NA,
                      se_slope = NA, ci_width = NA)

for(i in 1:length(sample_sizes)) {
  n <- sample_sizes[i]

  # Generate data
  x <- rnorm(n, 50, 10)
  y <- 2 + 0.5 * x + rnorm(n, 0, 5)

  # Correlation
  cor_result <- cor.test(x, y)

  # Regression
  model_temp <- lm(y ~ x)
  se_slope <- summary(model_temp)$coefficients[2, "Std. Error"]
  ci <- confint(model_temp)[2, ]
  ci_width <- ci[2] - ci[1]
}

```

```

# Store results
results$r[i] <- cor_result$estimate
results$p_value[i] <- cor_result$p.value
results$se_slope[i] <- se_slope
results$ci_width[i] <- ci_width
}

cat("Effect of Sample Size on Inference:\n\n")
print(round(results, 4))

cat("\nKey observation: As sample size increases,\n")
cat("  - Standard errors decrease\n")
cat("  - Confidence intervals narrow\n")
cat("  - Power to detect relationships increases\n")

```

6 Practice Problems

6.1 Problem 1: Cholesterol and Age

A study examines the relationship between age and total cholesterol in 25 adults.

```

# Data
set.seed(111)
age_chol <- seq(30, 70, length.out = 25)
cholesterol <- 160 + 1.2 * age_chol + rnorm(25, 0, 15)

```

Tasks:

- a) Create a scatter plot of cholesterol vs. age
- b) Calculate the Pearson correlation and test its significance
- c) Fit a linear regression model
- d) Interpret the slope in context
- e) Check regression assumptions
- f) Predict cholesterol level for a 50-year-old with 95% prediction interval

6.2 Problem 2: BMI and Blood Pressure

Examine the relationship between BMI and systolic blood pressure.

```
# Data
set.seed(222)
bmi <- rnorm(35, 27, 5)
sbp_bmi <- 95 + 1.8 * bmi + rnorm(35, 0, 8)
```

Tasks:

- a) Calculate correlation and test significance
- b) Fit linear regression
- c) Report the regression equation
- d) What is the predicted SBP for BMI = 30?
- e) Create a plot with regression line and 95% confidence bands

6.3 Problem 3: Comparing Correlations

Compare the correlation between height and weight for males vs. females.

```
# Data
set.seed(333)
height_male <- rnorm(25, 178, 8)
weight_male <- 0.9 * height_male - 75 + rnorm(25, 0, 6)

height_female <- rnorm(25, 165, 7)
weight_female <- 0.85 * height_female - 65 + rnorm(25, 0, 5)
```

Tasks:

- a) Calculate correlation for males
- b) Calculate correlation for females
- c) Compare the two correlations. Are they similar?
- d) Create side-by-side scatter plots
- e) Fit separate regression models for each group

7 Saving Your Work

IMPORTANT: Save your R script!

```
# At the top of your R script, add:  
# Name: Your Name  
# Date: Today's Date  
# Lab: Week 9 - Correlation and Simple Linear Regression  
  
# Save your workspace  
save.image("week9_lab.RData")  
  
# To load it later:  
# load("week9_lab.RData")
```

8 AI-Enhanced Learning Tips

Using AI Tools for Regression Analysis

Good prompts for AI assistance:

- “Explain the difference between correlation and regression with examples”
- “How do I interpret a regression slope of 1.25 mmHg per year?”
- “Create R code to check regression assumptions with diagnostic plots”
- “What does it mean if residuals show a curved pattern?”
- “Generate scatter plot with regression line and prediction bands”

AI can help with:

- Interpreting regression output
- Creating publication-quality plots
- Checking assumptions
- Troubleshooting R code errors
- Understanding confidence vs. prediction intervals
- Explaining statistical concepts

Remember:

- Always verify AI-generated results
- Check that interpretations make biological sense
- Examine plots visually, don’t just trust numbers
- AI is a learning tool, you are the analyst!

9 Key Takeaways

1. **Correlation:** Measures strength and direction of linear relationship (-1 to $+1$)
2. **Regression:** Models relationship and allows prediction
3. **Always visualize:** Scatter plots reveal patterns that statistics might miss
4. **Check assumptions:** Use residual plots (linearity, normality, constant variance)
5. **Interpret slope in context:** What does 1-unit change in X mean for Y?
6. R^2 : Proportion of variance explained (0 to 1)
7. **Prediction intervals:** Wider than confidence intervals (account for individual variation)

8. **Correlation $\neq$ causation:** Need experiments for causal claims

10 R Functions Reference

- **Correlation:**

- `cor(x, y)` – Correlation coefficient
- `cor.test(x, y)` – Test significance, get CI
- `cor(x, y, method="spearman")` – Spearman correlation

- **Regression:**

- `lm(y ~ x)` – Fit linear model
- `summary(model)` – Detailed output
- `coef(model)` – Extract coefficients
- `confint(model)` – Confidence intervals for coefficients
- `anova(model)` – ANOVA table

- **Prediction:**

- `predict(model, newdata)` – Point predictions
- `predict(model, newdata, interval="confidence")` – CI for mean
- `predict(model, newdata, interval="prediction")` – PI for individual

- **Diagnostics:**

- `plot(model)` – Diagnostic plots (residuals, Q-Q, etc.)
- `residuals(model)` – Extract residuals
- `fitted(model)` – Extract fitted values
- `shapiro.test(residuals(model))` – Test normality

- **Plotting:**

- `plot(x, y)` – Scatter plot
- `abline(model)` – Add regression line
- `lines(x, y)` – Add lines to existing plot

11 Additional Resources

- R Documentation: `?lm`, `?cor.test`, `?predict`
- Textbook: Pagano & Gauvreau, Chapters 17 and 18
- Quick-R: Linear Regression (<https://www.statmethods.net/stats/regression.html>)
- R for Data Science: <https://r4ds.had.co.nz/>